A Literature-Implied Countable-Compacta Obstruction for Gorak's Net Banach-Stone Problem

Math discovery packet

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1 Status

This is a lightweight *literature-implied answer*, not an original solution. The source paper is R. Gorak, *Coarse version of the Banach-Stone theorem*, arXiv:1005.0937. In the final remarks, Problem `problem Banach-Stone ogolny` asks whether, for compact spaces X, Y, the inequality

$$d_N(C(X), C(Y)) < 2$$

forces X and Y to be homeomorphic. The next problem asks for the answer when X, Y are countable compacta, and also asks what happens when X is the usual convergent sequence.

The supporting paper is A. Prochazka and L. Sanchez-Gonzalez, *Low distortion embeddings between $C(K)$ spaces*, arXiv:1408.0211.

2 Identification

The supporting paper recalls the Mazurkiewicz–Sierpinski classification: every countable Hausdorff compact space is homeomorphic to an ordinal interval of the form

$$[0, \omega^\alpha \cdot n], \qquad \alpha < \omega_1, \quad 1 \leq n < \omega.$$

Corollary `c:AnswerGorak` in arXiv:1408.0211 states that if $\gamma \neq \alpha < \omega_1$ and $m, n \in \mathbb{N}$, then

$$d_N(C([0, \omega^\gamma \cdot n]), C([0, \omega^\alpha \cdot m])) \geq 2.$$

Immediately after the corollary, the authors state that this partially answers Problem 2 in Gorak's paper.

Consequently, for countable compacta, net distance strictly below 2 forces equality of the ordinal exponent in the Mazurkiewicz–Sierpinski normal form. Equivalently, it rules out all countable compacta of different Cantor–Bendixson height in the threshold < 2 problem.

3 Scope

This does not settle Gorak's countable-compacta problem in full. The supporting paper itself notes that the same-exponent finite multiplicity question remains outside the stated corollary: the spaces

$$[0, \omega^\alpha \cdot m] \quad \text{and} \quad [0, \omega^\alpha \cdot n]$$

with $m \neq n$ are not separated by Corollary `c:AnswerGorak`.

In particular, for the source paper's test case where X is a convergent sequence, this packet records only the height obstruction. It does not prove that a countable compactum Y with $d_N(C(X), C(Y)) < 2$ must be homeomorphic to X.

4 Search Evidence

The run's lightweight indexes had no existing direct packet for arXiv:1005.0937 or this countable-compacta height obstruction. The identification comes from the local source of arXiv:1408.0211, whose introduction and Corollary `c:AnswerGorak` explicitly connect the result to Gorak's Problem 2.

5 References

- R. Gorak, *Coarse version of the Banach-Stone theorem*, arXiv:1005.0937.
- A. Prochazka and L. Sanchez-Gonzalez, *Low distortion embeddings between $C(K)$ spaces*, arXiv:1408.0211.
- S. Mazurkiewicz and W. Sierpinski, classical classification of countable compact Hausdorff spaces, as cited in arXiv:1408.0211.